

CRUDE AND PRODUCT TRANSPORT — TRUNK AND PRODUCT PIPELINES

Design Basis, Design Basis Accident, and Military Action Extension

DBA-OIL-FC4: Crude and Product Transport Facility Class

*Issued under DBA-OIL Sector Framework (v1.7) and DBA-EN-SF Energy Sector Framework
(v0.7)*

Version 0.3 — DRAFT

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Timothy E. Roxey

Eclectic Technologies

*For review by engineers, regulators, asset owners, and policy analysts responsible for
protecting energy sector critical infrastructure.*

Preface

The petroleum pipeline is the invisible artery of the American economy. It moves crude oil from wellheads to refineries and refined products from refineries to the terminals where consumers ultimately draw supply. It does this continuously, underground, across thousands of miles, without the visibility of a refinery stack or an offshore platform. When it works, no one notices. When it stops, everyone does — within days.

The Colonial Pipeline ransomware attack of May 2021 established a principle that this document takes as its analytical foundation: a sophisticated adversary does not need to physically destroy a pipeline to shut it down. An operator who cannot trust its own control systems will shut down voluntarily, because the alternative — running a hazardous liquid pipeline blind — is operationally and legally untenable. The six-day Colonial shutdown, driven entirely by an IT-side ransomware event with no damage to the pipeline itself, produced regional fuel shortages across the Eastern Seaboard. A state actor with the same objective and no constraint on kinetic action can produce the same supply disruption with a physical repair timeline measured in months rather than days.

Version 0.1 of this instrument established the DBA-MA scenarios for deliberate military action against crude and product trunk pipeline infrastructure: the cyber-kinetic product pipeline attack targeting Appalachian chokepoint pump stations, the multi-point kinetic rupture exceeding simultaneous repair capacity, and the compound attack on intersecting corridors with Hormuz closure amplification. It also established three pipeline-specific AI governance interaction points and five regulatory gap recommendations.

Version 0.3 completes the three-layer instrument by formalizing the non-adversarial DB layer — converting the v0.1 §3.1 narrative events into a structured event code table and adding the natural hazard and operational event categories that v0.1 omitted — and by establishing the Supply Delivery Duration (SDD) metric as the governing consequence metric for the pipeline transport facility class. SDD is a new series metric, proposed first at the FC4 level and warranting future sanctioning at the DBA-OIL sector framework and DBA-EN-SF levels in the same manner that PPSD was established at FC2 and sanctioned upward.

Section 1 — Introduction

1.1 Purpose

This document establishes the full three-layer Design Basis — Design Basis Accident — Military Action framework for the crude and product transport facility class. Version 0.1 established the DBA-MA scenarios directly, with a partial non-adversarial bounding event section in narrative form. Version 0.3 formalizes the DB layer into a structured event code table consistent with the series standard established in DBA-GAS-FC2 v0.2, adds the natural hazard and operational event categories omitted from v0.1, establishes the Supply Delivery Duration

(SDD) metric as the governing consequence metric, and develops the DBA consequence envelope with its Zone 1/2/3 acceptance criterion framework.

The addition of the formal DB layer is analytically necessary, not merely structural. The v0.1 MA scenarios — particularly Scenario A's voluntary shutdown dynamic — derive their consequence severity from the non-adversarial baseline: an operator who shuts down voluntarily on suspicion of SCADA compromise is making the same decision a prudent operator makes in response to a non-adversarial SCADA failure. The DBA-MA extension's force is precisely that it produces a consequence indistinguishable from a non-adversarial event — and that indistinguishability is the adversary's primary instrument of denial. A formal non-adversarial DB layer makes the comparison explicit and grounds the MA extension analytically.

1.2 Scope and Applicability

This document addresses crude and product trunk pipeline systems: large-diameter. These high-pressure liquid pipeline networks move crude oil from domestic producing regions and import terminals to refineries, and move refined petroleum products from refineries to distribution terminals. The facility class encompasses mainline pump stations, pipeline segments including water and environmentally sensitive crossings, SCADA and pipeline control systems, and associated measurement and regulation infrastructure.

Offshore export pipelines are addressed in DBA-OIL-FC2. Terminals and storage facilities are addressed in DBA-OIL-FC3. Natural gas transmission pipelines are addressed in DBA-GAS-FC1. Where FC4 pipeline events produce consequences that propagate to connected terminals or refineries, this document addresses those consequences as they originate at the pipeline without displacing the dedicated facility-class analysis for the receiving facility.

1.3 SDD — Governing Consequence Metric

Supply Delivery Duration (SDD) is established in this document as the governing consequence metric for the crude and product transport facility class. SDD is defined as the continuous interval of pipeline delivery curtailment — measured from the initiating event that reduces pipeline throughput below the committed delivery volume at the delivery point, to the restoration of full throughput capacity at that point.

SDD is a new series metric, first established at the FC4 level. It is the pipeline transport analog to PPSD (established at DBA-GAS-FC2 §1.3, sanctioned at DBA-EN-SF v0.6 §4.4) and SABD (established at DBA-ES-CntlCo v0.2 §1.2). All three metrics share the same structural logic: they measure the duration of functional consequence at the facility's primary delivery interface, use a Zone 1/2/3 escalation structure with T₁ and T₂ calibration thresholds, and terminate when delivery is confirmed restored to specification. SDD has been sanctioned at the DBA-OIL Sector Framework level (DBA-OIL-SF v1.7 §4.7) and at the DBA-EN-SF level (DBA-EN-SF v0.7 §4.4), consistent with the inductive sanctioning pathway established for PPSD.

SDD begins at the moment pipeline throughput at the delivery point drops below the minimum committed delivery volume or below the transmission system's minimum pressure floor at the receiving terminal or refinery. SDD ends at the moment full throughput capacity is confirmed restored and stable at the delivery point. The SDD interval includes SCADA reconstitution time, physical repair time, regulatory authorization time, and any environmental remediation hold that prevents restart — whichever governs the critical path.

1.4 Relationship to Governing Documents

This document is issued under the DBA-OIL Sector Framework (v1.7) and DBA-EN-SF: Energy Sector Design Basis Framework (v0.7). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. The Bright Line and Command Broker architecture is inherited from DBA-EN-SF and applied with pipeline-specific adaptation in §6. The Cross-Sector Consequence Elevation Rule (DBA-EN-SF §3.4) applies when the pipeline is the sole supply path for a Tier 2 or Tier 3 designated facility. The governing foundational instruments are FD-BL (placement), FD-BL-D1 (mode of discharge), and FD-BL-D2 (indication integrity), applied here by reference.

1.5 Position in Series

Document: DBA-OIL-FC4. Version: 0.2 — Full three-layer instrument.

Parent: DBA-OIL Sector Framework v1.7. Series Grandparent: DBA-EN-SF v0.7.

Gap closure: Completion of this instrument closes Gap 2.3 in the DBA-EN series gap register (DBA-OIL FC2/FC3/FC4 — DB Layer Only).

Section 2 — Pipeline Profile and Asset Taxonomy (Retained from v0.1)

The pipeline profile and asset taxonomy established in v0.1 §§2.1 through 2.5 are carried forward without substantive change. The governing text remains in v0.1. Where v0.1 and v0.2 conflict, v0.2 governs. A summary reference is provided below.

2.1 Asset Taxonomy Summary

Asset Class	Key Components	DB Bounding Concern	SDD Relevance
Crude trunk pipelines	Large-diameter mainline pipe, inline inspection equipment, block valves, cathodic protection systems	Pipeline rupture at a high-consequence area crossing — PHMSA regulatory restart timeline governs recovery beyond physical repair.	SDD: PHMSA restart authorization adds 2–12 months beyond physical repair at watercourse crossings

Product pipelines	Mainline pipe, batch interface zones, product quality measurement stations, manifold, and breakout terminals	Voluntary shutdown on SCADA compromise — entire system offline without physical damage	SDD Zone 1 begins at voluntary shutdown; Zone 3 if SCADA reconstitution exceeds T ₂
Pump stations	Centrifugal pump units, electric motor or turbine drivers, switchgear, local controls, SCADA RTU	Pump unit destruction at hydraulic chokepoint — 14–20 week procurement timeline, gates restart beyond SCADA reconstitution.	SDD: pump unit procurement is the governing critical path for Scenario A; RT_HVLLC analog for pipeline class
SCADA and control systems	Pipeline control center, RTUs at pump and valve stations, historian, batch tracking, CPM leak detection	Voluntary shutdown on SCADA compromise; false data injection masking rupture or generating spurious shutdown	SDD Zone 1 begins at SCADA-initiated shutdown; 4–6 week forensic reconstitution minimum
Environmentally sensitive crossings	Watercourse crossings (navigable and non-navigable), drinking water source crossings, and ecologically sensitive area segments	Rupture release at watercourse crossing — OPA 90 / PHMSA response triggers restart authorization process independent of physical repair timeline	SDD: environmental remediation hold can extend Zone 3 to 6–12 months per crossing beyond physical repair

2.2 Major Pipeline Systems — Consequence Profile (Retained from v0.1)

System	Type	Capacity (bpd approx.)	Consequence of Lost
Colonial Pipeline (Lines 1 & 2)	Product	~2,500,000	Eastern Seaboard gasoline, diesel, jet fuel — approximately 45% of regional supply— shortages in Southeast within 5

			days, Mid-Atlantic within 10 days, New England within 14 days.
Enbridge Mainline (Lines 1–6)	Crude	~3,000,000	Primary Canadian and Bakken crude supply to the Midwest and Gulf Coast refineries—refinery feedstock curtailment within days; no alternative crude routing at equivalent volume.
Dakota Access Pipeline	Crude	~750,000	Bakken crude supply to Gulf Coast markets. Loss forces Bakken producers to rail — higher cost, lower volume, longer transit.
Plantation Pipeline	Product	~600,000	Southeast product supply is parallel to Colonial. Partial redundancy with Colonial, but cannot absorb the full Colonial volume loss.
Seaway Pipeline	Crude	~850,000	Cushing, Oklahoma, to Gulf Coast refineries. Loss isolates Cushing crude storage from Gulf Coast refinery demand.

Section 3 — Design Basis: Non-Adversarial Bounding Events

3.1 DB Layer Purpose and Structure

The Design Basis layer establishes the cause-agnostic postulated events that a crude and product transport facility must be demonstrated to withstand. Version 0.1 §3.1 contained three non-

adversarial bounding events in narrative form. Version 0.3 formalizes those events into the series-standard event code table and adds the natural hazard and operational event categories that v0.1 omitted.

Four event categories organize the DB layer for the pipeline transport facility class: equipment failure (EF), natural hazard (NH), operational event (OE), and cross-sector cascade (XC). The EF events correspond to the three v0.1 §3.1 narrative events. NH and OE events are new in v0.2.

3.2 Equipment Failure (EF) Postulated Events

Code	Event	Initiating Condition	Governing Concern
EF-1	Pipeline rupture at a high-consequence area crossing	Mechanical failure of a pipeline segment at a navigable waterway, drinking water source, or populated area crossing — from corrosion, third-party damage, or pressure cycle fatigue. Large-volume liquid release to the surrounding environment.	The 2010 Enbridge Line 6B rupture in Marshall, Michigan, is the reference event: corroded pipeline segment failure at a wetland crossing producing an 843,000-gallon diluted bitumen release requiring four years and approximately one billion dollars to remediate. The PHMSA regulatory restart authorization process — pipeline isolation, environmental assessment, hydrostatic pressure testing, and inspection — produces a restart timeline that significantly exceeds the physical repair duration. At major watercourse crossings, the combined physical-plus-regulatory timeline can reach 6

			to 12 months per location.
EF-2	Pump station fire — major unit damage	Seal failure or high-pressure piping leak at a pump unit ignites; fire destroys the pump unit and damages the associated electrical switchgear and local control infrastructure. Pipeline throughput at the affected segment drops to zero or to available bypass routing capacity.	Pump station fire is the non-adversarial analog of the kinetic pump station attack in the MA extension. The repair timeline — pump unit procurement (14–20 weeks under normal conditions), installation, electrical infrastructure reconstruction, and PHMSA inspection — governs the SDD at the affected segment. The pump unit procurement lead time is the governing RT_HVLLC analog for the pipeline facility class.
EF-3	SCADA system failure — loss of pipeline control	A distributed control system or SCADA hardware failure removes automated pipeline control from one or more segments. Batch tracking is lost; product custody transfer cannot be verified. The pipeline must be shut down pending SCADA restoration.	SCADA failure is the non-adversarial analog of the voluntary shutdown event in the MA extension. Unlike a refinery DCS failure affecting a single facility, a trunk pipeline SCADA failure affects the entire system simultaneously. Manual operations on a major product pipeline are not feasible at normal throughput across the full right-of-way.

			SDD Zone 1 begins at shutdown initiation; SCADA reconstitution minimum is 4–6 weeks for a major system.
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3.3 Natural Hazard (NH) Postulated Events

Code	Event	Initiating Condition	Governing Concern
NH-1	Flooding — pump station electrical infrastructure submersion	Flood event submerges pump station electrical switchgear, MCC rooms, or substations, removing power and control capability from one or more pump stations. The affected pipeline segment loses pumping capacity.	Pump stations in the Gulf Coast, Midcontinent, and river corridor locations are subject to flooding from tropical weather events, river flooding, and flash flooding. Electrical infrastructure submersion requires inspection, testing, and potential equipment replacement before re-energization — the inspection and testing requirement, not physical flood damage, typically governs the restart timeline. A flood event affecting multiple pump stations simultaneously produces compound SDD across multiple pipeline segments.
NH-2	Freeze event — exposed aboveground piping at pump	Prolonged extreme cold produces ice formation in exposed	Pump stations in northern pipeline corridors —

	stations	aboveground pump station piping, instrument connections, and valve actuators. Loss of instrument function removes SCADA telemetry from affected stations; frozen valve actuators prevent emergency isolation.	Enbridge Mainline in Minnesota and Wisconsin, Dakota Access through the Dakotas — operate aboveground piping and instrumentation that is exposed to extreme cold. Freeze events that disable multiple stations simultaneously produce compound SDD across the corridor. The Winter Storm Uri (February 2021) experience across multiple infrastructure types is the governing empirical anchor for the compound freeze scenario.
NH-3	Seismic event — pipeline integrity at river crossings	Ground motion at a major river crossing produces stress on the pipeline at the bend point where it transitions from horizontal to angled crossing geometry. Coating damage, joint separation, or buckling may result. An integrity assessment is required before restarting at any crossing in the affected seismic zone.	River crossing segments are the most mechanically constrained sections of a major pipeline: they are typically directionally drilled beneath the waterway at depth, with bend radii and wall thickness specifications that are already at design limits. Seismic ground motion at a crossing that produces any detectable pipe movement triggers a PHMSA-required integrity assessment before restart. For

			crossings where a release would reach navigable water, the assessment standard is the same as EF-1, producing a restart timeline of weeks to months beyond the assessment period.
NH-4	Tropical weather event — Gulf Coast pipeline segment damage	Hurricane or tropical storm produces storm surge, wind damage, or debris impact at aboveground pump station infrastructure along the Gulf Coast pipeline corridor. Landfall segment damage may include pipeline exposure from shoreline erosion at coastal crossings.	Gulf Coast pipeline infrastructure — the origin end of Colonial Pipeline, Plantation, and the crude trunk lines connecting the Gulf Coast refinery complex — is concentrated in the zone of maximum hurricane impact. A major hurricane landfall along the upper Texas or Louisiana coast affects multiple pipeline systems simultaneously. The compound event — storm surge flooding of multiple pump stations plus potential coastal crossing exposure — produces the most severe natural hazard SDD scenario for the pipeline facility class.

3.4 Operational Event (OE) Postulated Events

Code	Event	Initiating Condition	Governing Concern
OE-1	Pipeline pig run — stuck intelligent pig	An in-line inspection tool (intelligent pig)	Stuck pig events are among the most

		becomes lodged in a pipeline segment during a scheduled inspection run. The pipeline must be depressurized and the pig extracted through an excavation at the lodgment point. Segment throughput is reduced or eliminated during extraction.	operationally disruptive non-adversarial events for major pipeline systems. A stuck intelligent pig in a major crude trunk or product pipeline requires segment isolation, excavation, and pig extraction before normal throughput can resume. For deep-buried river crossing segments where the pig becomes lodged at the crossing geometry, extraction may require directional drilling or underwater excavation, extending the SDD from hours to weeks.
OE-2	Batch interface mismanagement — product contamination	Incorrect batch sequencing or interface management at a product pipeline manifold produces commingling of incompatible product grades — gasoline commingled with jet fuel, or high-sulfur distillate commingled with low-sulfur diesel. The contaminated batch must be isolated, downgraded, and diverted before delivery continues.	Product contamination events on major product pipelines are consequential because they affect the product quality at every delivery terminal downstream of the contamination point until the interface is isolated. A contamination event on Colonial Pipeline affects the entire Northeast delivery system from the contamination point to Linden, New Jersey. Product

			quality testing and interface management before any delivery can resume produces SDD at all affected terminals simultaneously.
OE-3	Pressure transient — rapid valve closure	Emergency valve closure at a pump station or block valve station produces a pressure wave (water hammer) that travels along the pipeline. If the pressure transient exceeds the pipeline MAOP at any point, automatic pressure relief valve actuation or pipeline shutdown is required. PHMSA notification is required for any pressure exceedance above MAOP.	Water hammer events from rapid valve closure are a known operational risk for high-velocity liquid pipeline operations. A pressure transient that exceeds MAOP at any location triggers a PHMSA notification and integrity assessment before full pressure restart is authorized. The governing concern is the compound event: a pressure transient at a location coinciding with a pre-existing corrosion anomaly that the last intelligent pig inspection did not flag. The transient precipitates a failure at the anomaly location, converting an OE event into an EF-1 class consequence.
OE-4	Control room staffing below the minimum complement	Pipeline control room operator staffing falls below the minimum qualified controller	PHMSA Control Room Management regulations (49 CFR Part 195.446) require

		complement required to maintain all required pipeline control functions simultaneously, due to illness, failure to report, or extended emergency exhausting shift rotation capability.	pipeline operators to establish and maintain minimum staffing levels for control room operations. Below-minimum staffing events require the operator to assess whether safe operations can be maintained and to restrict operations or shut down if they cannot. Extended emergency conditions — a multi-day weather event or a compound infrastructure disruption — can produce staffing shortfalls through a combination of controller fatigue and access denial, both of which are non-adversarial DB events with the same consequence structure as the human layer events in DBA-ES-CntlCo.
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3.5 Cross-Sector Cascade (XC) Postulated Events

Code	Event	Source	FC4 Consequence
XC-1	Electric supply loss to pump stations — electric-drive equipment	Electric sector (DBA-ES) event producing supply loss to pump station electric motor drivers along a pipeline corridor.	Electric-drive pump stations on the affected corridor lose pumping capacity simultaneously. Gas-turbine-drive stations continue operating. The XC-1 event

			produces SDD from a non-petroleum-sector initiating condition — the gas-electric bidirectional cascade pathway extended to the petroleum sector through shared electric supply infrastructure.
XC-2	Refinery throughput curtailment — crude demand reduction	DBA-OIL-FC1 event (refinery major outage) reduces crude pipeline demand at the refinery delivery point. Crude trunk pipeline throughput must be reduced; upstream crude storage and production allocation adjustments are required.	XC-2 runs in the reverse direction from the primary consequence chain: a refinery event reduces pipeline utilization rather than the pipeline event curtailing refinery supply. The SDD consequence is upstream — crude wellhead production curtailment or Cushing storage buildup as the pipeline operator reduces throughput to match reduced refinery demand.
XC-3	Terminal storage overflow — pipeline delivery constraint	DBA-OIL-FC3 event (terminal storage capacity exhaustion or outage) prevents product delivery at the pipeline terminus. The pipeline must reduce throughput or shut in as the terminal cannot accept further product.	XC-3 bridges the pipeline and terminal facility classes through the storage constraint pathway. A terminal fire or flooding event that takes tankage out of service can fill remaining storage within hours, forcing pipeline delivery shutdown even though the pipeline

			itself is undamaged. SDD from XC-3 is governed by the terminal restoration timeline, not the pipeline repair timeline.
XC-4	NGL pipeline outage — petroleum sector cascade from gas processing	DBA-GAS-FC2 event (processing plant outage) producing NGL pipeline curtailment that affects product pipeline batch sequencing or butane blending supply at pipeline injection points.	XC-4 is the petroleum-sector manifestation of the NGL consequence chain identified in DBA-GAS-FC2 §2.2. NGL curtailment affects gasoline blending (butane content) and can force batch sequencing adjustments on product pipelines that blend NGLs at injection points. The SDD consequence is a product quality constraint rather than a volume constraint.

Section 4 — Design Basis Accident: Non-Adversarial Consequence Envelope

4.1 Bounding Event Identification

The DBA layer identifies the bounding events from the DB layer that define the consequence envelope within which protective measures and recovery architecture must perform. For the pipeline transport facility class, three bounding events govern across the equipment failure, natural hazard, and operational event categories.

Bounding Event	Source Event	Governing Consequence	Metric
Pipeline rupture at a major watercourse crossing — PHMSA regulatory hold	EF-1	Product or crude release to a navigable waterway. PHMSA regulatory	SDD: physical repair 4–12 weeks; regulatory hold adds 2–12 months.

		authorization timeline (2–12 months beyond physical repair) governs SDD at the crossing segment regardless of how quickly the physical repair is completed.	Governing single-segment SDD bounding event.
SCADA system failure — full pipeline shutdown	EF-3	Full pipeline system offline; batch tracking lost; product custody transfer cannot be verified. SCADA reconstitution minimum 4–6 weeks before restart is authorized.	SDD Zone 1 through Zone 3, depending on system size and forensic complexity. Governing voluntary-shutdown-class SDD event.
Compound pump station flood — multiple stations simultaneously	NH-1 compound	Multiple pump stations on the same corridor simultaneously lose power and control capability. Corridor throughput drops to zero. Sequential restart as each station's electrical inspection is completed.	SDD: compound event extending across multiple segments simultaneously—duration governed by sequential electrical inspection and testing timeline.

4.2 SDD — Supply Delivery Duration Metric Definition

Supply Delivery Duration (SDD) measures the continuous interval of pipeline delivery curtailment at the governing delivery point. For crude trunk pipelines, the governing delivery point is the refinery inlet flange or the pipeline terminal custody transfer meter. For product pipelines, the governing delivery point is the terminal receipt meter at the first downstream terminal receiving less than the committed volume.

SDD is distinguished from physical repair duration by the inclusion of all intervening steps that prevent restart after physical repair is complete: SCADA forensic investigation, PHMSA regulatory authorization, environmental remediation holds, and product quality testing following

contamination events. The SDD clock does not stop when physical repairs are complete — it stops when full throughput is confirmed restored and stable at the delivery point.

SDD is not additive across pipeline segments. For a single pipeline system with multiple simultaneous events, the governing SDD is the longest single-segment SDD in the critical path from the supply source to the delivery point. If Segment A has a 4-week SDD and Segment B (downstream of A) has a 6-month SDD, the governing system SDD is 6 months — not 6 months plus 4 weeks — because delivery cannot resume until the critical-path segment is restored.

4.3 Consequence Taxonomy

Tier	Pipeline Facility Consequence Definition	Bounding Event	Governing Metric
Tier 1	Facility-level consequence: single pipeline segment throughput reduction. The downstream delivery point receives reduced volume, but not below the minimum contracted delivery. A terminal or refinery can absorb the reduction through storage drawdown or throughput adjustment without curtailing downstream product supply.	EF-2 (single pump station fire, non-chokepoint location), OE-1 (stuck pig, short-segment isolation), OE-3 (pressure transient, MAOP exceedance without release)	SDD Zone 1: segment-level curtailment, system-level delivery maintained above minimum threshold.
Tier 2	Grid-level consequence: pipeline delivery below the minimum contracted volume at the governing delivery point. The downstream terminal or refinery cannot fully absorb the reduction; product	EF-1 (rupture at non-major crossing), EF-3 (SCADA failure — full system), NH-1 (single pump station flood), OE-2 (product contamination — corridor-wide)	SDD Zone 2: system-level curtailment confirmed. Alternative supply mechanisms are activated.

	supply to the regional market is impacted. Alternative routing, storage drawdown, or marine imports must be activated to offset the curtailment partially.		
Tier 3	Cross-sector cascading consequence: pipeline delivery curtailment produces refinery feedstock shortfall or regional product supply crisis. Military fuel logistics were affected. International supply context amplification is compounded by global disruption. Cross-sector consequence elevation rule (DBA-EN-SF §3.4) triggered.	EF-1 at major watercourse crossing (6–12 month regulatory hold), EF-2 at hydraulic chokepoint (pump unit procurement timeline), NH-4 (Gulf Coast hurricane compound event), XC compound events	SDD Zone 3: Tier 3 cascade developing. Federal Coordination Threshold notification. IEA coordination evaluated.

4.4 SDD Acceptance Criterion Framework

The SDD acceptance criterion framework establishes the Zone 1/2/3 escalation structure for pipeline delivery curtailment events. T₁ and T₂ calibration values are pipeline-system-specific and are established in the site-specific design basis implementation.

SDD Zone	Condition	Required Response	Escalation Trigger
Zone 1 (0 to T ₁)	Pipeline throughput is reduced, but delivery at the governing point is above the minimum contracted volume. System-level delivery maintained—storage drawdown at	Pipeline operator: initiate repair or restoration sequence. Receiving terminal: activate storage drawdown procedures. Downstream shippers: notified,	Delivery at governing point confirmed below minimum contracted volume, or storage drawdown at receiving terminal projected to exhaust within T ₂ interval.

	receiving terminal within normal operating range.	monitoring supply position.	→ Zone 2.
Zone 2 (T ₁ to T ₂)	Delivery below the minimum contracted volume. Receiving terminal or refinery drawing storage below the normal operating minimum. Alternative supply mechanisms activated: marine imports, Plantation Pipeline diversion, and SPR drawdown assessment.	PHMSA notification (if release involved). TSA notification (if cyber component suspected). DOE ESF-12 monitoring activated—alternative supply routing formalized. Jones Act waiver assessed for marine supplementation.	Regional product supply shortage confirmed at the retail level, or refinery feedstock curtailment confirmed, or military fuel logistics affected. → Zone 3.
Zone 3 (beyond T ₂)	Regional product supply crisis or refinery feedstock curtailment confirmed. Military fuel logistics were affected. The international supply context has degraded. SPR drawdown committed.	Federal Coordination Threshold notification. DOE ESF-12 activation. NSC coordination for compound events. IEA emergency mechanism evaluated—mandatory demand allocation assessed. Cross-sector consequence elevation rule (DBA-EN-SF §3.4) applied.	Zone 3 is the maximum consequence condition for the pipeline transport facility class—no further escalation zone.

T₁ calibration guidance: T₁ is the terminal storage buffer duration — the period for which the downstream terminal or refinery can maintain operations above the minimum supply threshold using on-hand storage before requiring supplementary supply. For product terminals, T₁ is typically 5 to 14 days, depending on storage capacity and demand draw rate. For refineries receiving crude trunk supply, T₁ is typically 3 to 10 days of crude storage capacity at design throughput.

T₂ calibration guidance: T₂ is the alternative supply bridge duration — the period for which marine imports, alternative pipeline routing, and storage drawdown can offset the pipeline curtailment before retail or military supply shortages develop. For the Eastern Seaboard market served by Colonial Pipeline, T₂ is bounded by the marine import ramp-up timeline and the rate at which Southeast terminal storage exhausts ahead of marine supply arrival.

4.5 RT_HVLLC Application — Pump Unit as Governing Long-Lead Component

The RT_HVLLC metric applies to the pump unit as the governing high-value long-lead component for the pipeline transport facility class. RT_HVLLC for a major centrifugal pump unit at a chokepoint pump station is 14 to 20 weeks under normal supply chain conditions, reflecting both manufacturing time and the limited number of manufacturers producing large centrifugal pumps for high-pressure liquid pipeline service at specific flow and pressure configurations.

RT_HVLLC for the pump unit is the governing physical recovery constraint for Scenario A (kinetic attack on chokepoint pump stations) and the compound Scenario C event. Unlike the gas processing cold box (8–16 months), the pump unit RT_HVLLC of 14–20 weeks means that SCADA reconstitution (4–6 weeks) is not the critical path for kinetic attack scenarios — the pump unit procurement is. This distinction governs the Phase 1 recovery action: pump unit procurement inquiry must be initiated within hours of the attack, not after SCADA forensics are complete.

Section 5 — DBA-MA Military Action Extension (Retained from v0.1, Updated for v0.2)

5.1 MA Extension Foundation

The DBA-MA extension applies the four MA characteristics — intentionality, persistence, precision, and denial — to the non-adversarial bounding events established in §§3 and 4. The three scenarios from v0.1 are carried forward without substantive change to their analytical content. Version 0.3 adds SDD zone references to the consequence envelopes and formally grounds each scenario in the DB layer events it exploits.

The governing insight from v0.1 remains: the pipeline adversary's most powerful instrument is the voluntary shutdown dynamic. An operator who suspects SCADA compromise shuts down before any kinetic action occurs — and that voluntary shutdown is the consequence equivalent of a physical attack on a pipeline that cannot be operated manually at scale. The MA extension adds kinetic pump station destruction to extend the SDD beyond the SCADA reconstitution window, converting what would be a Zone 2 SCADA event into a Zone 3 physical recovery event.

5.2 Four MA Characteristics Applied to Pipeline Transport Facility Class

MA Characteristic	Application	Governing Concern
Intentionality	Adversary selects pump stations at hydraulic chokepoints — terrain-constrained locations where loss of pumping capacity stops all flow through the	The chokepoint pump station is the pipeline analog of the cold box in DBA-GAS-FC2: the single component whose loss stops system operation and whose replacement

	affected segment with no bypass option. Target selection exploits public-domain knowledge of pipeline corridor hydraulics: elevation profiles, river crossing locations, and pump station spacing are available in public FERC filings and pipeline operator tariff submissions.	timeline governs the SDD. The adversary's intelligence requirement for chokepoint identification is low — the hydraulic constraints that make a pump station a chokepoint are the same constraints that make it appear in FERC filings as a capacity limitation.
Persistence	Adversary maintains SCADA pre-positioning for a period sufficient to map the pipeline control architecture, stage manipulation commands, and identify the optimal timing window for the combined cyber-kinetic execution. Persistence also applies post-attack: adversary interdicts pump unit procurement to extend SDD Zone 3 duration beyond the non-adversarial RT_HVLLC baseline.	SCADA pre-positioning on major pipeline systems has been documented in classified threat reporting for over a decade. The persistence characteristic means that the adversary's access may precede any observable indicator by months or years. Detection of pre-positioning before execution is the only intervention point that prevents the voluntary shutdown dynamic from initiating.
Precision	Adversary injects SCADA data at specific measurement points to produce a false pipeline state — a simulated rupture signature at a location chosen to trigger precautionary shutdown — while the kinetic component targets the chokepoint pump stations. The false rupture signature is designed to be indistinguishable from EF-1 until the kinetic strikes execute.	Precision in the pipeline context operates at two levels: the precision of the false SCADA signature (designed to trigger the specific shutdown protocol that takes the entire system offline) and the precision of the kinetic targeting (pump stations at hydraulic chokepoints whose loss prevents restart even after SCADA reconstitution). The compound precision of both vectors simultaneously is the MA contribution above the non-adversarial baseline.

Denial	MA-1 class attack: SCADA manipulation produces a false system state that the operator acts on voluntarily — the operator's own safety response becomes the adversary's instrument of initial denial. Post-kinetic denial: the adversary understands that the operator's first recovery action is pump unit procurement, and can target the supply chain to extend the SDD Zone 3 duration.	Denial in the pipeline context has three phases: initial denial through voluntary shutdown (SCADA manipulation), sustained denial through physical damage (chokepoint pump station destruction), and extended denial through supply chain interdiction (pump unit procurement obstruction). Each phase exploits a different vulnerability, and each phase's duration adds directly to the SDD Zone 3 clock.
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5.3 Scenario A — Cyber-Kinetic Attack on Major Product Pipeline (Retained from v0.1)

Coordinated cyber-kinetic attack against Colonial Pipeline targeting the Appalachian crossing pump stations. The SCADA component injects anomalous pressure readings mimicking a rupture signature to trigger precautionary voluntary shutdown. The kinetic component executes within 30 minutes: precision UAS strikes against three Appalachian crossing pump stations, destroying primary pump units, electrical switchgear, and local control infrastructure.

DB layer grounding: Scenario A exploits EF-3 (SCADA failure producing voluntary shutdown) as its initiating mechanism and converts it into EF-2 (pump station destruction) as its persistence mechanism. The adversary uses the non-adversarial voluntary shutdown protocol as the opening phase of a kinetic attack — the operator's correct response to EF-3 becomes the first phase of the MA consequence.

SDD consequence: Zone 1 begins at voluntary shutdown ($T=0$). Zone 2 is confirmed within T_1 as the kinetic strikes are confirmed, and the SCADA forensic investigation begins. Zone 3 as pump unit procurement timeline (14–20 weeks) extends beyond SCADA reconstitution (4–6 weeks), and the governing SDD extends to 5–8 months — the pump unit delivery and installation critical path. Federal Coordination Threshold notification is an immediate $T=0$ action for the kinetic component.

5.4 Scenario B — Multi-Point Kinetic Rupture of Major Crude Trunk Pipeline (Retained from v0.1)

Coordinated kinetic attacks against a major crude trunk pipeline at four to six locations chosen to exceed the pipeline operator's simultaneous repair capacity and to include at least two

environmentally sensitive crossings where PHMSA regulatory restart requirements impose a timeline significantly longer than the physical repair.

DB layer grounding: Scenario B exploits EF-1 (pipeline rupture at high-consequence area crossing) at multiple simultaneous locations. The adversary's contribution above EF-1 is simultaneity — four to six simultaneous ruptures that exceed the repair crew and equipment queue capacity, converting independent EF-1 events into a compound scenario whose recovery timeline is governed by the queue, not by any individual event.

SDD consequence: Zone 3 immediately for a four-to-six simultaneous rupture scenario — the compound event exceeds regional response capacity and produces refinery feedstock curtailment within days. The governing SDD is the longest regulatory restart timeline across all affected crossing segments — 6 to 12 months for major watercourse crossings.

5.5 Scenario C — Compound Attack on Intersecting Pipeline Corridors (Retained from v0.1)

Coordinated attacks against both a major crude trunk pipeline and a major product pipeline sharing a common corridor or serving the same regional market, executed within a 48-hour window. Scenario A modality against the product pipeline; Scenario B modality against the crude trunk line. Timed to coincide with a Hormuz closure or Persian Gulf supply disruption that degrades import alternatives and SPR availability.

DB layer grounding: Scenario C compounds EF-1 and EF-2/EF-3 simultaneously across two pipeline systems. The MA contribution is the elimination of the cross-system redundancy that each pipeline normally provides for the other — a consequence that no non-adversarial compound event achieves because non-adversarial events are not deliberately targeted to remove both systems' mutual offset capability simultaneously.

SDD consequence: Zone 3 immediately. The governing SDD is the longer of Scenario A (5–8 months) and Scenario B (6–12 months), running in parallel with no mutual offset. Under the Hormuz closure context, import alternatives are constrained, and the SPR drawdown is partially committed, extending the effective SDD consequence beyond the physical pipeline recovery timeline.

Section 6 — Dual Design Basis Architecture and AI Governance (Retained from v0.1, Updated for v0.2)

The three AI governance interaction points established in v0.1 §3.3 are retained. Version 0.3 adds a fourth interaction point derived from the DB layer: the OE-4 control room staffing event and its human layer analog to the CntlCo HL events, with implications for Gen AI operator support governance.

Interaction Point	Bright Line Position	Requirement
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Pipeline SCADA and predictive control — pressure management	Below the Bright Line for deterministic pressure management within the trained envelope	SCADA pressure management outputs must be range-checked against independent pressure sensor data before pump commands execute. Anomalous pressure trajectories trigger automatic pump shutdowns independent of SCADA command confirmation. SCADA operating band validation is architecturally isolated from the corporate IT network.
Anomaly detection and CPM leak detection ML	Below the Bright Line for signal classification within the trained envelope	CPM systems must receive sensor data through paths architecturally independent of the SCADA command path. Anomalous sensor patterns inconsistent with normal operations trigger investigation before shutdown rather than automatic shutdown. CPM model integrity is validated against an adversarial input scenario set at each model update cycle.
Pipeline scheduling and optimization — Gen AI	Above Bright Line — non-deterministic	Command Broker is mandatory for any pipeline operating parameter change recommended by the Gen AI scheduling tool. Command Broker must be a qualified pipeline controller; a software component cannot execute a function. Manual scheduling capability is maintained independently of Gen AI tools.
Control room Gen AI support under OE-4 staffing conditions (NEW in v0.2)	Above Bright Line — non-deterministic; elevated risk under reduced staffing	When control room staffing is at or below the minimum qualified complement (OE-4 condition), Gen AI operator support tools present elevated governance risk: a fatigued or cognitively overloaded controller is less able to

		exercise effective Command Broker judgment over non-deterministic Gen AI output. The Design Basis requires that Gen AI operator support tools automatically flag when the control room is below minimum staffing complement, and that the Command Broker authorization workflow for Gen AI recommendations requires an additional confirmation step under below-minimum-staffing conditions.
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Section 7 — Protective Action Thresholds (Retained from v0.1, Updated for v0.2)

The protective action thresholds, threat indicator taxonomy, decision authority framework, and federal coordination thresholds established in v0.1 §5 are retained. Federal coordination thresholds are updated to reference SDD Zone 3 as the governing federal notification criterion.

7.1 Federal Coordination Thresholds — SDD Integration

Threshold	Trigger Condition	Notification Requirement
SDD Zone 3 — regional supply crisis	Regional product supply shortage confirmed at the retail level, refinery feedstock curtailment confirmed, or military fuel logistics affected as a consequence of pipeline SDD.	DOE ESF-12 activation. NSC for compound events. Cross-sector consequence elevation rule (DBA-EN-SF §3.4) applied.
Kinetic event at pump station or right-of-way	Confirmed kinetic strike at any pump station or pipeline segment of a major crude trunk or product pipeline	FBI, DOD (if state-actor attribution suspected), CISA, TSA, PHMSA, DOE ESF-12.
SCADA intrusion — nation-state attribution	Confirmed SCADA intrusion with suspected nation-state attribution at any major pipeline operator	CISA, FBI, DOD. Parallel notification to sector peer pipeline operators through ISAC. TSA Security Directive notification within

		12 hours.
Pipeline release to navigable waterway	Release to navigable waterway from any cause consistent with adversary action above the PHMSA significant incident threshold	PHMSA, USCG Federal On-Scene Coordinator, EPA, state environmental agency. OPA 90 response framework activated.
Compound event — simultaneous multi-system	Simultaneous anomalous SCADA events or kinetic events at pump stations on two or more major pipeline systems within 48 hours	All of the above plus: NSC, State Department for transboundary systems, IEA notification if the global supply context is degraded.

Section 8 — Recovery Architecture (Retained from v0.1, Updated for v0.2)

The recovery architecture established in v0.1 §6 is retained. Version 0.3 adds SDD Zone transition milestones and clarifies the pump unit procurement as a Phase 1 action, not Phase 2.

8.1 Recovery Phase — SDD Zone Alignment

Recovery Phase	Timeline	SDD Zone Target	Governing Constraint
Phase 1 — Immediate	0–72 hours	Prevent Zone 3; achieve Zone 2 characterization	Emergency pipeline shutdown and pressure relief; environmental release containment and agency notification. SCADA forensic investigation initiated—physical damage assessment at the attacked pump stations. Pump unit specification and procurement inquiry initiated — Phase 1 action because procurement timeline gates physical recovery. DOE ESF-12 notification. Alternative supply

			routing activation: Plantation Pipeline, marine import authorization, and SPR drawdown assessment.
Phase 2 — Short-term	72 hours to 8 weeks	Sustain Zone 2; prevent further cascade	SCADA forensic investigation continuing. Physical damage assessment complete. Pump unit procurement is underway. Environmental remediation initiated. Alternative supply arrangements formalized. PHMSA coordination for restart authorization planning.
Phase 3 — Medium-term	8 weeks to SDD expiry	Zone 1 on partial capacity; full capacity at RT_HVLLC expiry	SCADA reconstitution completed. Pump unit installation at affected stations. PHMSA inspection and restart authorization per segment. Sequential segment restart. Supply chain monitoring for adversary interdiction of pump unit components. SDD clock stops when full throughput is confirmed restored and stable at the governing delivery point.

8.2 SCADA Reconstitution as Recovery Prerequisite

A pipeline that has experienced a confirmed or suspected SCADA intrusion cannot restart under automated pipeline control until SCADA forensic investigation and reconstitution are complete. The minimum 4 to 6 week reconstitution timeline cannot be compressed without compromising the forensic integrity required for attribution and for the operator's own assurance that the intrusion has been fully remediated. For Scenario A events where the kinetic phase extends the SDD beyond the SCADA reconstitution window, SCADA reconstitution should be completed. The control system is declared trustworthy before pump unit installation work begins at the kinetically damaged stations — ensuring that when physical repairs are complete, the pipeline can restart under a validated control environment without an additional SCADA reconstitution delay.

Section 9 — Gaps and Recommendations (Retained from v0.1, Updated for v0.2)

The five regulatory gap recommendations from v0.1 §7 are retained. Version 0.3 adds a sixth gap derived from the DB layer: the absence of SDD as a recognized metric in any federal pipeline emergency response framework.

Gap	Description	Status	Lead Action
G-1	TSA mandatory cybersecurity directives for pipeline operators do not address physical security. Pump stations at hydraulic chokepoints — the highest-consequence physical attack target for the pipeline facility class — have no mandatory physical security assessment requirement.	Open	TSA should develop mandatory physical security standards for pipeline pump stations and control centers, using the Colonial Pipeline vulnerability assessment framework as the regulatory model.
G-2	No federal or industry program pre-positions critical spare pump units for major pipeline systems. The 14–20 week pump unit procurement timeline is the governing	Open	DOE, in coordination with TSA and PHMSA, should assess the feasibility of a strategic spare pump unit program for hydraulic chokepoint stations on major pipeline

	physical recovery constraint for Scenario A and compound Scenario C events.		systems. The electric sector STEP program provides the analytical model.
G-3	No federal emergency response framework addresses the compound consequence of simultaneous crude trunk and product pipeline disruptions affecting the same regional market during a global supply disruption event.	Open	DOE should lead development of a compound petroleum supply disruption response framework integrating domestic pipeline recovery, IEA coordination, and military fuel priority allocation.
G-4	The Enbridge Mainline and other Canada-U.S. transboundary crude pipeline systems have no bilateral pipeline security annex addressing threat information sharing, joint vulnerability assessment, and coordinated emergency response.	Open	A bilateral pipeline security annex under the Canada-U.S. Critical Infrastructure Protection framework should be developed before a compound transboundary event creates reactive conditions for poorly designed bilateral arrangements.
G-5	No domestic energy emergency response plan explicitly addresses the compound consequence of a Hormuz closure coinciding with a major domestic pipeline attack. The two response	Open	DOE, in coordination with NSC and IEA, should develop a compound scenario planning baseline establishing SPR drawdown priorities and IEA coordination protocols for simultaneous international and

	mechanisms — IEA stock drawdown and SPR drawdown — have not been stress-tested against simultaneous activation.		domestic activation.
G-6 (NEW in v0.2)	SDD (Supply Delivery Duration) is established as the governing consequence metric for the pipeline transport facility class in this instrument. No federal pipeline emergency response framework — PHMSA, TSA, DOE ESF-12 — currently uses a duration-based pipeline delivery curtailment metric equivalent to SDD. Consequence reporting under existing frameworks is event-based rather than duration-based, obscuring the supply impact of extended pipeline outages relative to brief disruptions.	Open	DOE ESF-12 and PHMSA incident reporting frameworks should incorporate SDD as a reporting metric for significant pipeline disruption events, enabling consequence-duration tracking that informs both emergency response decision-making and infrastructure investment prioritization. SDD has been sanctioned at the DBA-OIL Sector Framework level (DBA-OIL-SF v1.7 §4.7) and at the DBA-EN-SF level (DBA-EN-SF v0.7 §4.4). The sanctioning gap (G-6) is closed as of this revision.
0.3	June 2026	T. Roxey	Housekeeping corrections and one substantive status update. (1) Parent document references updated: DBA-OIL Sector Framework v1.6 → v1.7; DBA-EN-SF v0.6 → v0.7,

			in cover lineage lines, §1.4 issued-under sentence, and §1.5 parent/grandparent line. (2) SDD sanctioning status updated: “SDD warrants future sanctioning at the DBA-OIL Sector Framework level and at DBA-EN-SF in a future revision” corrected to reflect that sanctioning is complete: DBA-OIL-SF v1.7 §4.7 and DBA-EN-SF v0.7 §4.4 both now carry SDD as a sanctioned series-standard metric. Gap G-6 sanctioning sub-item closed. The G-6 regulatory pathway gap (SDD not recognized in the federal pipeline emergency response framework) remains open.
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Section 10 — References

Regulatory and Standards References

U.S. Department of Transportation, Pipeline and Hazardous Materials Safety Administration. Pipeline Safety Regulations — Hazardous Liquid Pipelines. 49 CFR Part 195. Washington, D.C.: PHMSA. Current version.

U.S. Department of Transportation, Pipeline and Hazardous Materials Safety Administration. Control Room Management. 49 CFR Part 195.446. Washington, D.C.: PHMSA. Current version.

Transportation Security Administration. Pipeline Security Directives (SD-02A, SD-02B, SD-02C). Washington, D.C.: TSA. Current versions.

U.S. Environmental Protection Agency. Oil Pollution Act of 1990 (OPA 90) and Spill Prevention, Control, and Countermeasure Rule. 40 CFR Part 112. Washington, D.C.: EPA. Current version.

U.S. Coast Guard. National Contingency Plan — Federal On-Scene Coordinator. 40 CFR Part 300. Washington, D.C.: USCG/EPA. Current version.

Government and Policy References

U.S. Department of Transportation, Pipeline and Hazardous Materials Safety Administration. "Enbridge Incorporated Hazardous Liquid Pipeline Rupture and Release, Marshall, Michigan, July 25, 2010." Washington, D.C.: NTSB Pipeline Accident Report PAR-12-01, July 2012.

U.S. Department of Homeland Security, Cybersecurity and Infrastructure Security Agency. "Colonial Pipeline Cyber Incident — After-Action Review." Washington, D.C.: CISA, 2021.

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Menon, E. Shashi. Liquid Pipeline Hydraulics. New York: Marcel Dekker, 2004. (Pump station hydraulics and chokepoint analysis.)

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Series Parent and Cross-Reference Documents

Roxey, Timothy E. DBA-EN-SF: Energy Sector Design Basis Framework. Version 0.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Timothy E. DBA-OIL: Petroleum Sector Framework. Version 1.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Timothy E. DBA-OIL-FC1: Petroleum Refining. Version 1.3 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Timothy E. DBA-OIL-FC3: Crude and Product Terminals and Storage. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Timothy E. DBA-GAS-FC2: Natural Gas Processing. Version 0.3 — DRAFT. Eclectic Technologies, June 2026. (PPSD metric structure; series precedent for SDD Zone framework.)

Roxey, Timothy E. DBA-ES-CntlCo: Electric Sector Control-Company Functional Class. Version 0.3 — DRAFT. Eclectic Technologies, June 2026. (SABD metric structure; series precedent for functional consequence metric with Zone escalation framework.)

Roxey, Timothy E. RT_HVLLC Strategic Reserve White Paper. Version 0.1 — DRAFT. Eclectic Technologies, June 2026. (Pump unit strategic spare program — G-2 policy case.)

Version 0.3 — DRAFT Page

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. DBA-MA layer. Three scenarios: Scenario A (cyber-kinetic attack on major product pipeline — Colonial template with Appalachian chokepoint pump station destruction), Scenario B (multi-point kinetic rupture of major crude trunk pipeline — exceeding simultaneous repair capacity), Scenario C (compound attack on intersecting corridors with Hormuz closure amplification). Three AI governance interaction points. Five regulatory gap recommendations. Partial non-adversarial bounding events in §3.1 (narrative form, three events). Parent document: DBA-OIL Sector Framework v1.5.
0.2	June 2026	T. Roxey	Full three-layer instrument. DB layer formalized (§3): v0.1 §3.1 narrative events converted to structured event code table; natural hazard (NH) and operational event (OE) categories added; cross-sector cascade (XC) category added. Supply Delivery Duration (SDD) established as governing consequence metric for the pipeline transport facility class (§4.2) — new series metric, first established at FC4 level. DBA layer formalized (§4): bounding events identified, Tier 1/2/3 consequence taxonomy applied, SDD Zone 1/2/3 acceptance criterion framework established. MA extension (§5) retained from v0.1 with SDD reference integration. Parent document reference updated to DBA-OIL Sector Framework v1.6 and DBA-EN-SF v0.6. Gap 2.3 in the DBA-EN series gap register closed.